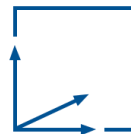


Communication Interfaces for Distributed Augmented Reality Applications in Industrial Contexts

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Final: Bachelor Informatics: Games Engineering

Supervisor: Prof. Gudrun Klinker, Ph.D

Introduction / Motivation

- Overall goal: digital twin of industrial facilities
- Problem: initially created CAD models get outdated [Rud+22]
- Solution: create 1:1 digital reconstruction of individual parts within the facility and integrate them into the CAD model [RK21]
- Solution approach: “3D Reconstruction” app to create Volumes of Interest (Vol)

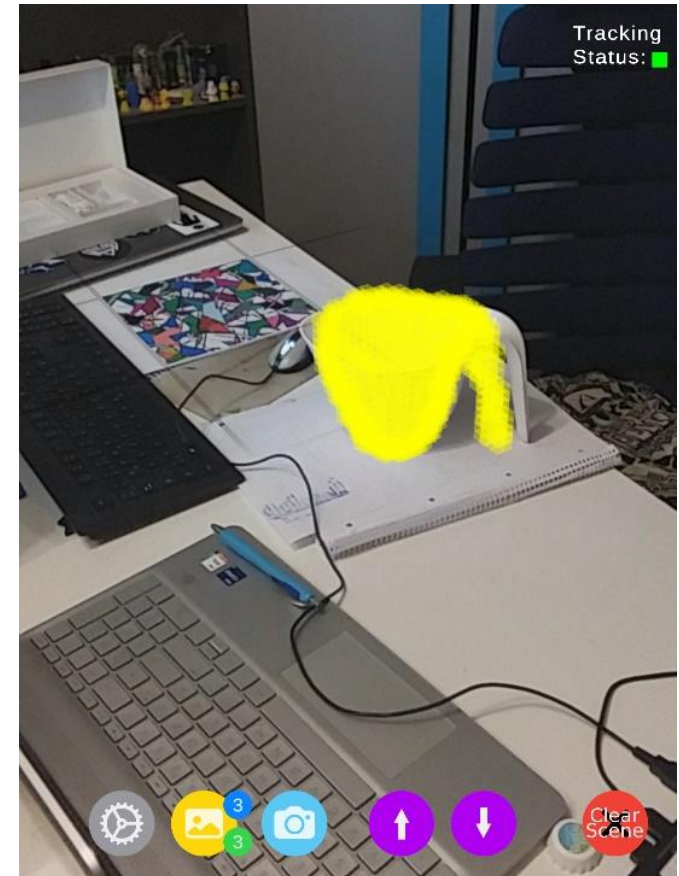
Introduction / Motivation

Previous state of the app [Sti22] [Klu23]:

- Image capture
- Image annotation
- Voxel carving
- Vol augmentation in the scene

Problem Description: Issues

- Vol can not be shared
- Representation of Vol is not ideal
 - Consists of small cube prefabs
 - In many cases too opaque
 - Needs more storage than necessary



Previous Volume visualization of the app

Goals of this Thesis

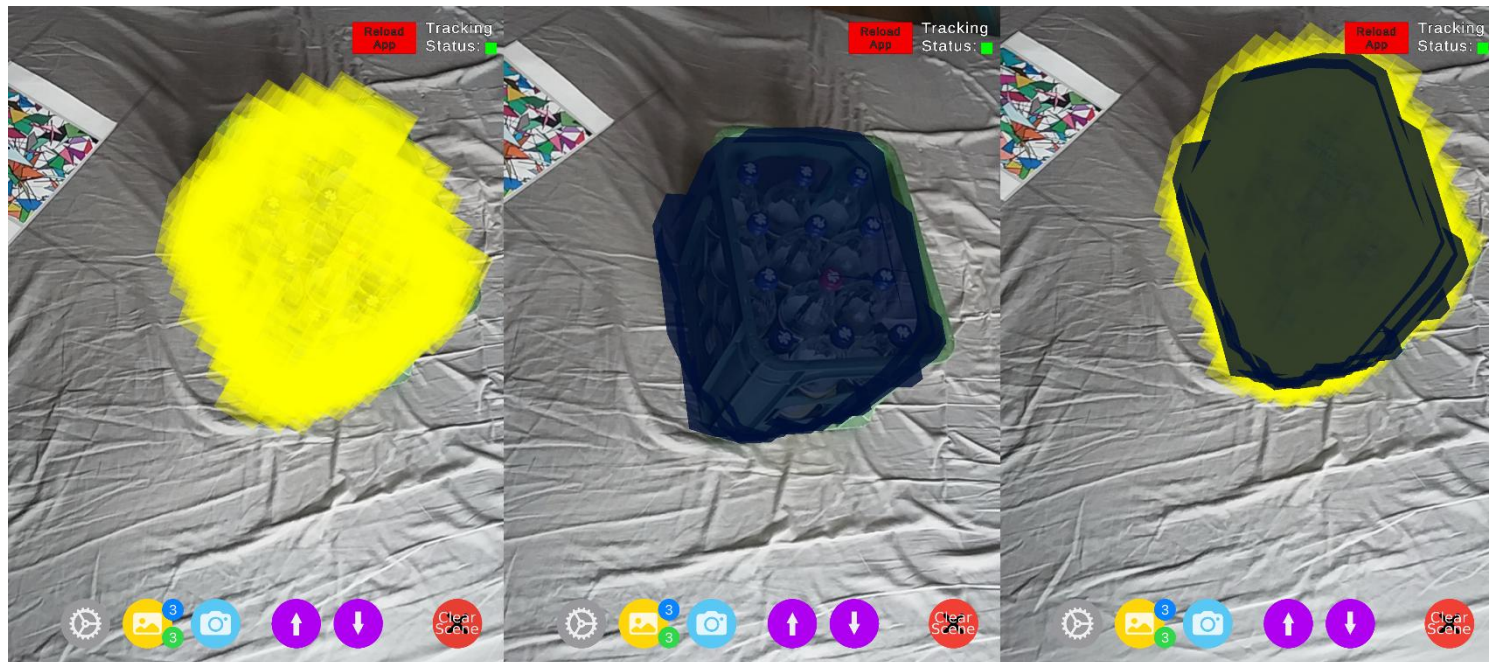
- Improve the visualization of the Vol
- Store the Vol
- Implement a communication interface to send the Vol and its pose to a server
- Add UI features proposed by [KLU23]
 - Portrait mode
 - Tracking status indicator
 - Restructure settings
 - Automated voxel size
- Investigate possible differences in usability test scores

Proposed Work / Approach

- Marching Cubes algorithm
- glTF im- & export
- Localhost python test server
- Preliminary user study

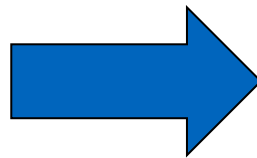
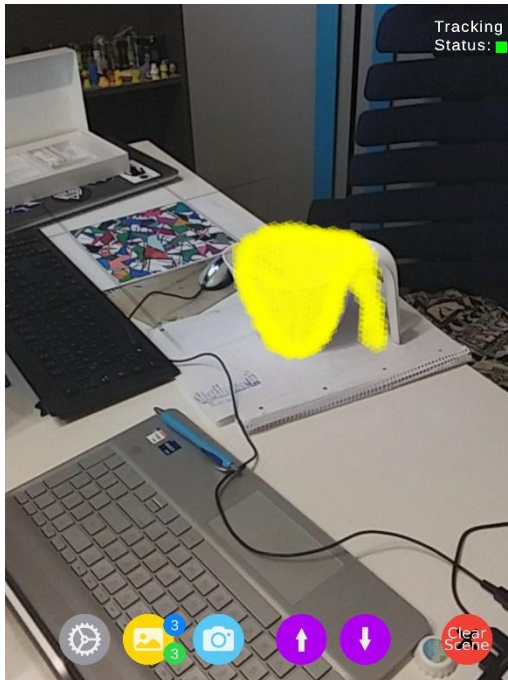
Implementation

- New voxelcarving
 - Check voxel center instead of eight corners
 - 3D boolean array instead of Vector3 list to store voxels



Implementation

- Marching Cubes algorithm
 - Plugin by Justin Hawkins [Haw22]
 - Adapter was needed



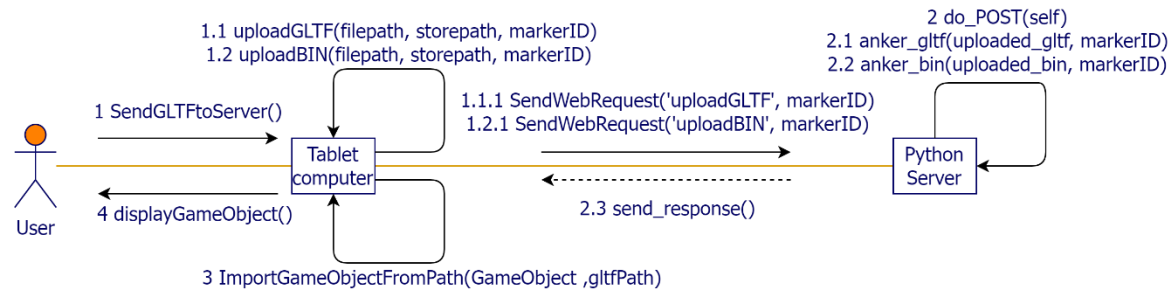
Implementation

- glTF im - & exporter
 - Plugins by Robert Long [Lon23] and Andreas Atteneder [Att22]
 - Creates .gltf and .bin file from Marching Cubes mesh
 - Creates gameobject with mesh from file

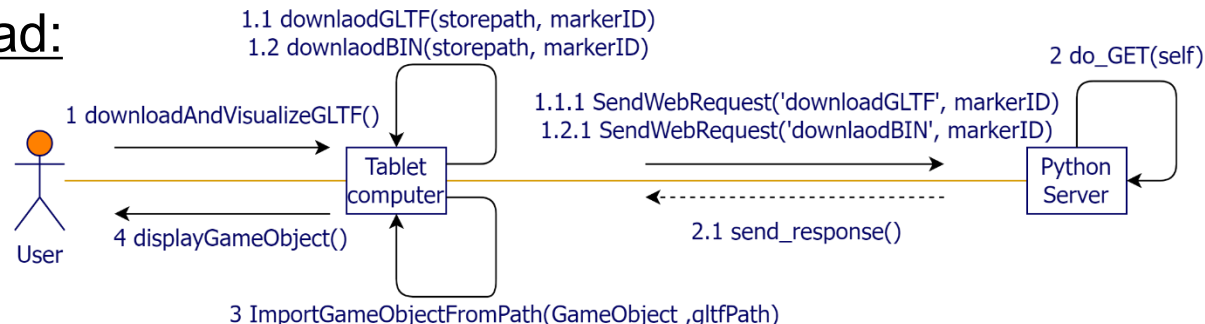
Implementation

- Testserver
 - Local host python server
 - Tablet and server connection via mobile phone hotspot

Upload:



Download:



Implementation

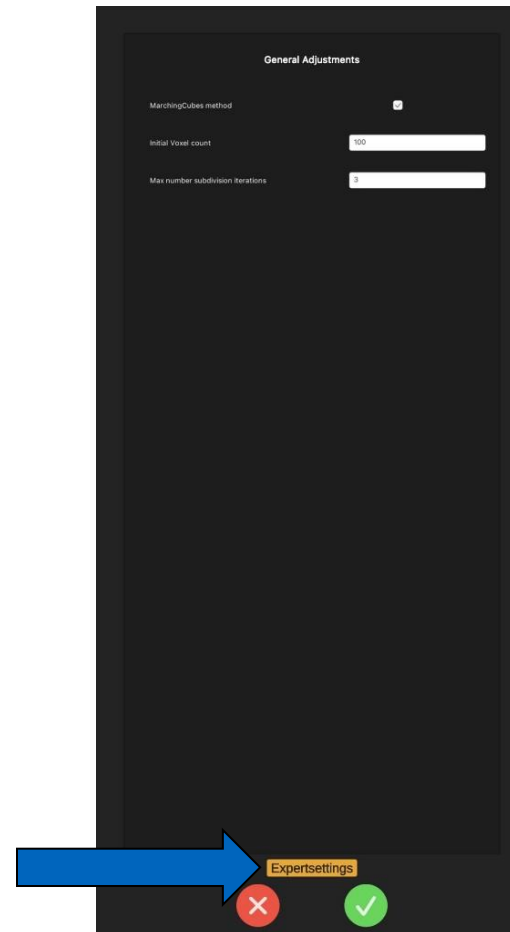
- Testserver



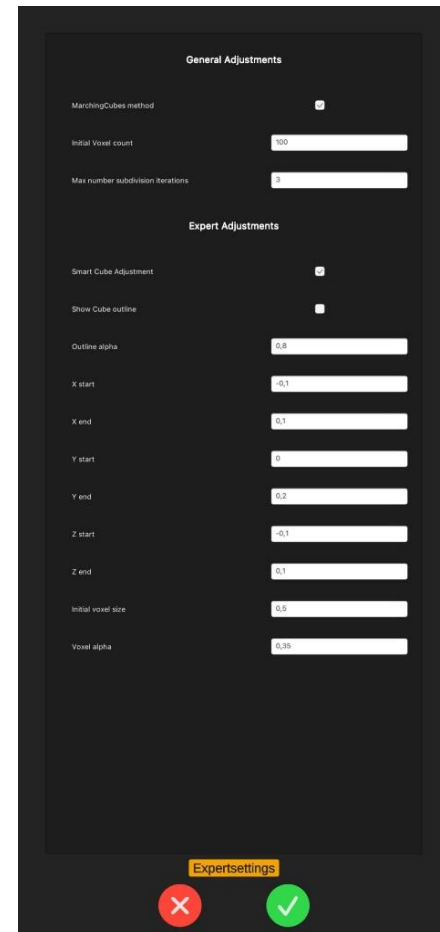
Scene with server answer

Implementation

- Settings tab
 - 3 main settings
 - Hidden „Expertsettings“



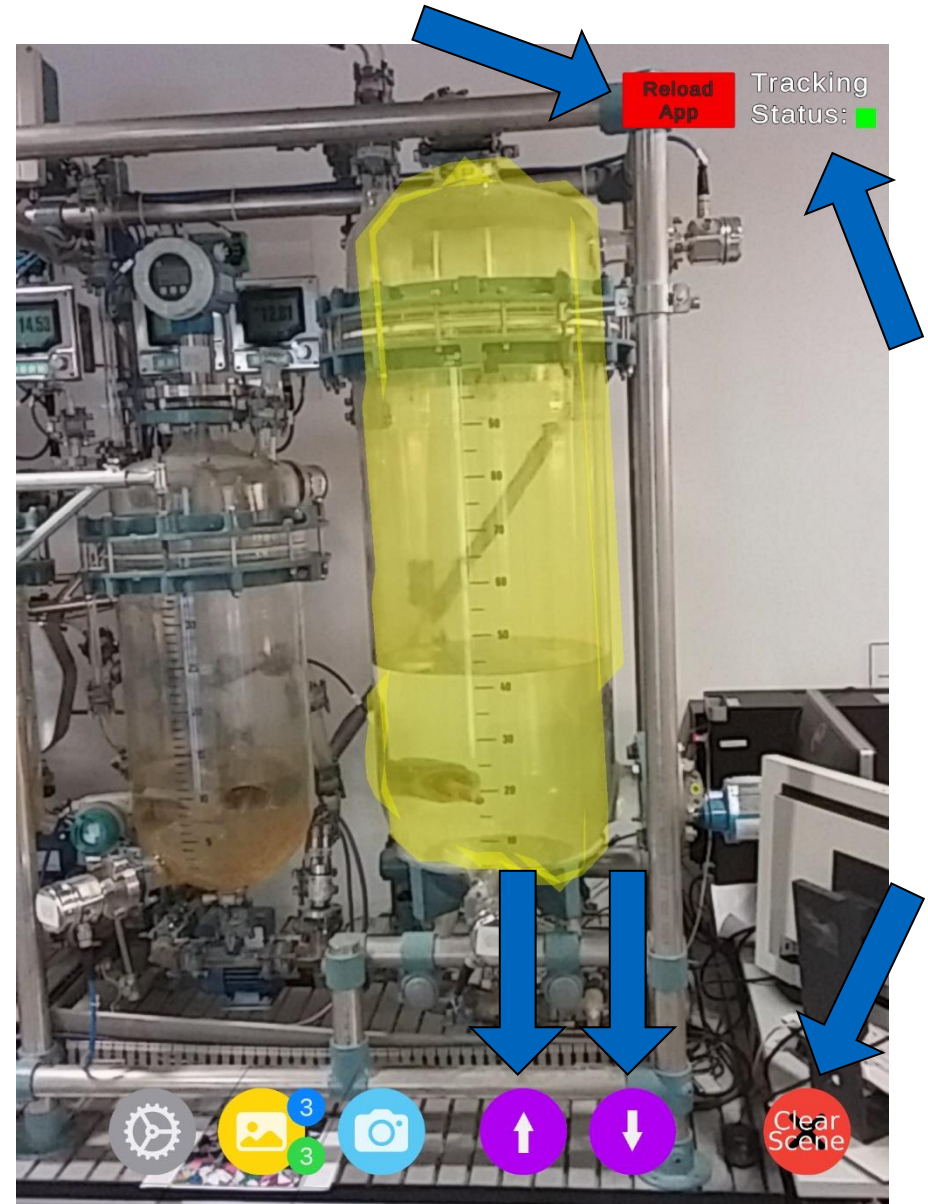
Normal Settings view



Clicked on „Expertsettings“

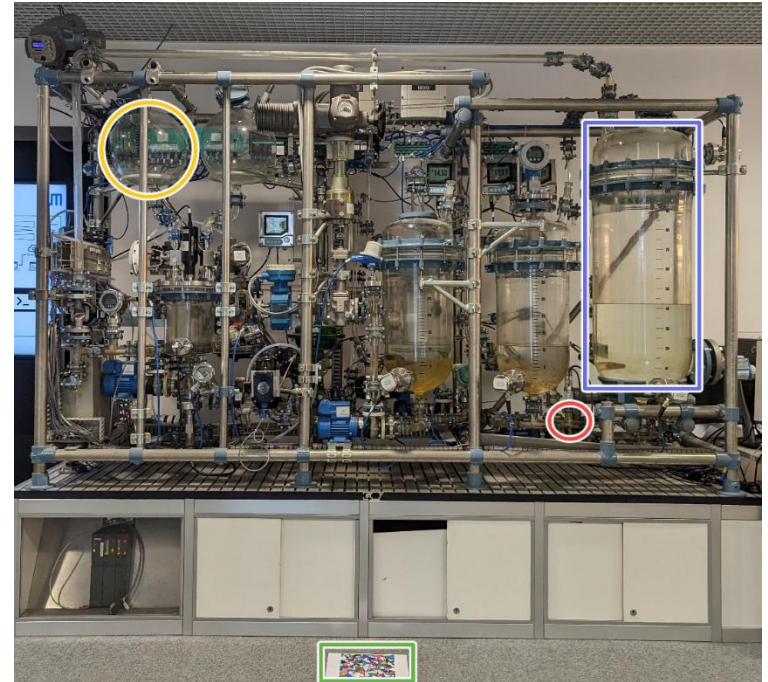
Implementation

- User interface elements



Evaluation (User Studies)

- Small study with 3 participants from TUM & LMU
 - System Usability Scale
 - NASA Task Load Index
 - Application Survey [KLU23]
 - Interview [KLU23]



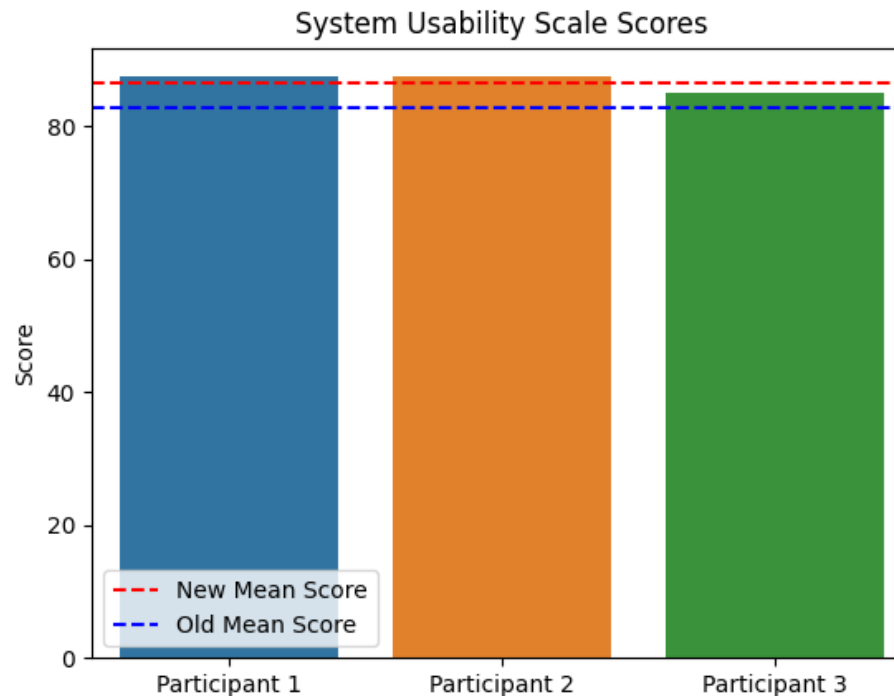
Test machine with marker and 3 objects of interest

Evaluation (User Studies)

- General observations
 - Confidence levels were 7 and 9 out of 10 to use the app
 - Difficulties with small objects

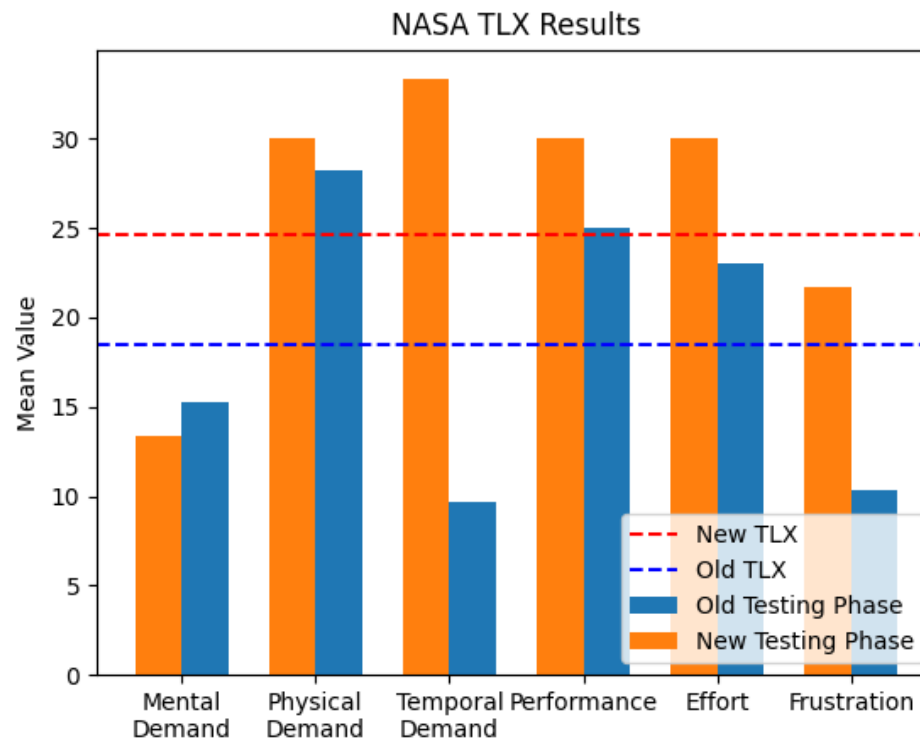
Evaluation (User Studies)

- System Usability Scale



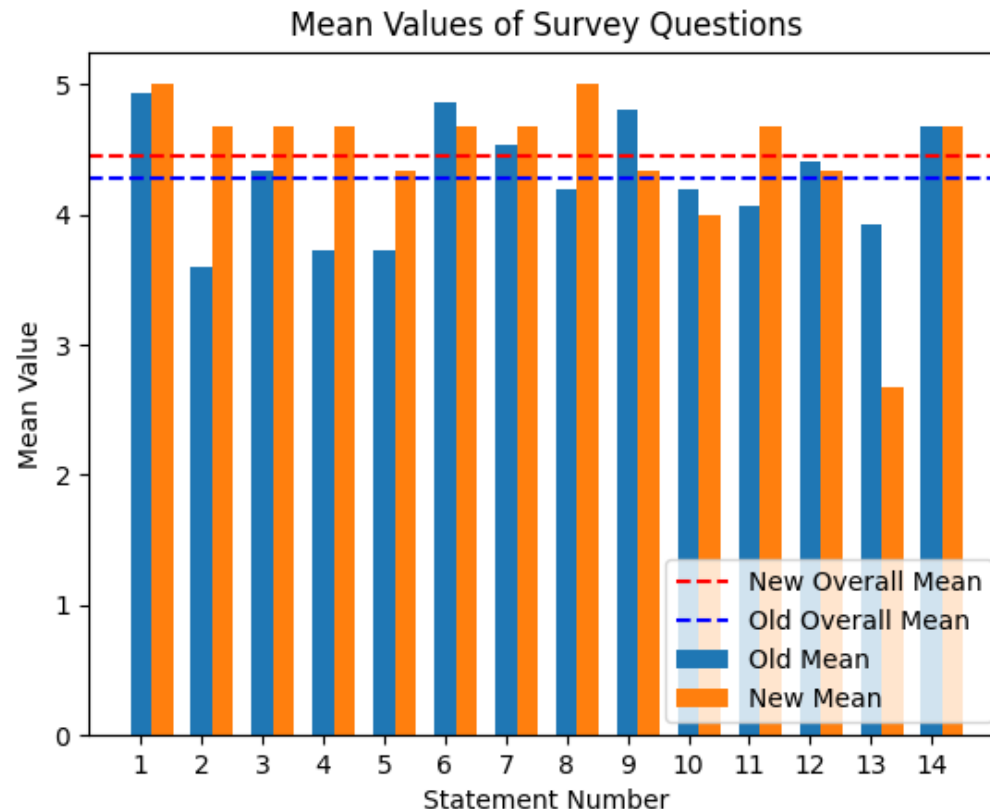
Evaluation (User Studies)

- NASA Task Load Index
 - Big difference in temporal demand → participants knew time was being measured



Evaluation (User Studies)

- Application Survey



Evaluation (User Studies)

- Interview
 - Suggested features by participants
 - Object Detection
 - Automated Initial Voxelcount

Discussion / Suggested Future Work

- Test scores have to be validated with larger study
- Reconstruction process for small objects (size detection?)
- Explore different button positions & devices
- Multi marker – multi Vol environment support

Conclusion

- Visualization of Vol with Marching Cubes
- Vol storage & sharing with glTF
- Test scores are promising but have to be validated

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